

Sunya Afrasiabi

email | personal website | linkedin

Education

University of Pennsylvania , BSE in Mechanical Engineering, BA in Philosophy	Sept 2024 - May 2028
• Relevant Coursework: Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Computer-Aided Design (CAD), Computer Programming (Python), Linear Algebra, Differential Equations, Probability	
University at Buffalo , Gifted Math Program	Sept 2023 - April 2024
• Relevant Coursework: College Calculus III, Introduction to Linear Algebra	

Experience & Leadership

Munch Industries, Inc. , Mechanical Engineer (Volunteer)	July 2026 - Present
• Design and fabricate the awning and structural framework for Orble, an automated boba tea vending machine startup, using SolidWorks, laser cutting, 3D printing, and PVC fabrication.	
University of Pennsylvania – Li Group , Undergraduate Researcher	May 2026 - Present
• Research insect-wing-inspired morphing structures, modeling compliant geometries in Rhino and Grasshopper to study parametric mechanical behavior. • Translate computational designs into prototypes using powder-based 3D printing for experimental validation.	
NASA L'SPACE Proposal Writing & Evaluation Experience , Radiation Subteam Lead	Sept 2025 - Dec 2025
• Developed innovative solution to the limitation of radiation-resistant materials in HDPE composites to enhance material performance under lunar radiation conditions • Gained experience in technical proposal composition, project evaluation, and NASA's mission review process. • Ranked in the Top 6 out of 40+ national teams.	
Penn Astronomical Student Association , Outreach & Marketing Director	Sept 2025 - Present
• Lead outreach initiatives and external partnerships, securing \$200 in sponsorship funding. • Configure and calibrate equatorial telescopes and optical tracking systems for public observation events.	
#MyStory Program , Founder and Project Manager	Jan 2021 - Aug 2024
• Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions. • Secured over \$24,000 via targeted donations, fundraisers, and corporate grant programs (including Riley's Way Foundation, Prudential, Ashoka, and the Peter & Elizabeth Tower Foundation). • Directed operations for 20+ community partners and 40+ volunteers.	

Technical Projects

Multi-Phase Butane Rocket & Numerical Analysis	Spring 2026
• Designed and optimized a variable-mass butane-powered water rocket using aerodynamic modeling, thermodynamic analysis, and experimental validation to navigate constrained flight paths and minimize target error. • Developed a custom Python Euler integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics; implemented trajectory reconstruction and multi-variable optimization to improve predictive accuracy.	
Mars Rover Chassis Prototype	Spring 2026
• Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures. • Performed FEA-driven design iterations to optimize load paths, bracket placement, wall thicknesses, and lightweight geometries while balancing structural integrity and mass reduction.	
Vacuum-Powered Agricultural Seed Cleaner	Fall 2025
• Designed and fabricated a vertical pneumatic seed-chaff separator with a zig-zag separation column, custom swirl chamber, laser-cut acrylic panels, and 3D-printed nozzles to optimize airflow behavior and separation efficiency.	

Papers

Linear Algebra Foundations of Neural Networks , University of Pennsylvania	August 2025
• Wrote expository research paper connecting linear algebra, neural networks, and robotics. (Link)	
Aviation Solutions for Wildlife Prevention , Wharton Undergraduate Aerospace Club	May 2025
• Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing \$6.4B+ market opportunities and potential \$5B annual cost savings. (Link)	

Skills

Software: Rhino 3D, Grasshopper, SolidWorks, MATLAB, Python, NumPy, FEA
Fabrication: Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly
Additional: Computational Geometry, Bio-Inspired Design, Technical Writing, Logic, Project Management